Bengaluru-based QpiAI open sources quantum SDK on GitHub

On July 7, 2026, QpiAI officially released its quantum software development kit (SDK) as open source software, making it globally available to developers, researchers, universities, and startups. The SDK repository is publicly hosted and available on GitHub to be directly integrated into existing development workflows.

The QpiAI quantum SDK is a Python-based SDK providing clean interfaces for circuit creation, simulation, algorithm development, and workflow execution. The SDK serves as a direct bridge to connect local workflows to QpiAI's actual physical hardware, specifically their 8-qubit and 25-qubit quantum computers, via the cloud execution platform, QpiAI-QCloud. It ships with built-in local state-vector and density matrix simulators.

This allows developers to fully test, prototype, and validate their quantum algorithms locally on their own machines before spending credits or running them on actual quantum hardware. In addition to standard quantum programming, the framework is specifically designed to support AI-assisted and agentic development workflows, helping automate the path from idea to implementation.

The open source release is strategically aligned with the objectives of India's National Quantum Mission to foster an indigenous, highly competitive quantum software talent pool. "India is entering a defining decade for quantum technologies, and open source software will be critical to building the talent, research, and innovation base that national leadership requires," said founder and CEO Dr Nagendra Nagaraja.

systems in sectors like automation, aerospace, defence, medical devices, and critical infrastructure. A headline feature is the fully automated generation of a software bill of materials (SBOM) during the system build process.

Requiring zero manual configuration, it logs all packages, files, and dependencies in the industry-standard SPDX v3 JSON format. This workflow, alongside automated security auditing tools based on French ANSSI guidelines and default GCC Fortification Level 3 runtime protections, is specifically designed to help manufacturers comply with the European Cyber Resilience Act (CRA). The platform introduces a graphical configuration wizard and over 400 precompiled software packages, enabling engineers to deploy lean Linux images with a minimal memory footprint and a significantly reduced attack surface.

Built on the Linux 6.12 Long-Term Support (LTS) kernel, ELinOS 8 guarantees up to ten years of standard security maintenance. It provides a unified development environment across Intel, Arm, and RISC-V architectures, backed by new Board Support Packages (BSPs) from major vendors like NXP, Toradex, TQ-Systems, and AMD Xilinx. Crucially, vendor BSP updates have been decoupled from the core product release cycle, allowing developers to accelerate hardware deployments without altering core project stability.

"ELinOS 8 is our most security-focused release to date. With integrated SBOM generation, ANSSI hardening validation, reproducible builds, Linux 6.12 LTS, and expanded BSP support, we are helping customers address growing cybersecurity and compliance requirements while accelerating development," said David Engraf, head of product development at SYSGO.

Linux Foundation launches Akrites to secure critical open source software

The Linux Foundation has launched Akrites, bringing together leading technology companies to protect critical open source software from AI-enabled cyber threats through a unified vulnerability response framework.

The initiative establishes a shared Security Incident Response Team (SIRT) and a standardised Coordinated Vulnerability Disclosure (CVD) process to identify, coordinate, remediate and responsibly disclose vulnerabilities before attackers can exploit them.

The move comes as frontier AI models are now capable of identifying software vulnerabilities within minutes, compressing the time between discovery and potential exploitation. Akrites promotes upstream-first patching, responsible disclosure and the use of established security standards including CVE, CWE, CVSS, EPSS, SSVC, VEX and TLP.

A key feature of Akrites is its ability to serve as a "maintainer of last resort" for critical open source projects without active maintainers, ensuring security patches continue to reach users while preserving maintainer control wherever possible. The framework also replaces fragmented security responses with a single trusted disclosure channel, reducing duplicate reports and conflicting patches.

Founding members include Amazon Web Services, Anthropic, Cisco, Google, IBM, JPMorganChase, Microsoft, GitHub, Nvidia, OpenAI, Red Hat and several